SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)
DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING
(AFFILIATED TO JNTUK, KAKINADA (RECOGNISED BY ALL INDIA COUNCIL FOR TECH. EDN., NEW DELHI)
(Accredited by NAAC, NBA NEW DELHI)
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Dr. B.R.K.Varma,
M.E.,Ph.D.
Professor &Head

06-07-2024

CIRCULAR

This is to inform all the members that the 11th Board of Studies (BOS) meeting of the Department of EEE, SRKR Engineering College, Bhimavaram will be held on **20-07-2024 (Saturday) at 10:00 AM** in PG Simulation Laboratory of the department. All the BOS members are invited to attend the meeting.

Agenda:

1. To discuss and finalize the scheme and syllabi for the Second year of R23 B. Tech (EEE) course curriculum.
2. Any other item with the permission of the chairman.

Head of the Department &
BOS Chairman
Head of EEE Department
S.R.K.R. Engg.College
BHIMAVARAM –534 204

Copy to

1. The members of BOS
2. The Principal, SRKREC
3. The Dean, Academics
4. Office file

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)
Accredited by NAAC with 'A+' Grade
Recognised as Scientific and Industrial Research Organisation
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

Date: 20-7-2024

Members of Board of Studies

S. No	Category	Name	Position	Signature
1	Chairman	Dr. B.R.K.Varma	Professor & Head Dept. of EEE SRKR Engineering College China Amiram, Bhimavaram-534204	20/7/24.
2	Expert Nominated by Vice-Chancellor	Dr. K.Venkata Reddy	Professor of EEE, UCEK JNTUK, Kakinada-533003	Attended in online mode
3	Subject Experts from outside Parent Universities	Dr. M. Damodara Reddy	Professor, Department of EEE College of Engineering, Sri Venkateswara University Tirupati-517502	Attended in online mode
4		Dr. Ch. V.V.S. Bhaskara Reddy	Professor, Dept. of EEE, College of Engineering, Andhra University, Visakhapatnam-530003.	Attended in online mode
5		Dr P.Sankar	Assistant Professor Dept. of EEE, NIT AP, Tadepalligudem, -534102, AP	Attended in online mode
6	Representative from Industry/Corporate Sector/Allied Area Relating to Placement	Akula Ramesh	Principal Engineer, Mercedes Benz Research and Development India, Bangalore.	Attended in online mode
7	Postgraduate Meritorious Alumnus	Dr.G.Kasi Viswanadha Raju	Lead Specialist, Power System Consulting, Vestas Technology R&D, Chennai-600119	Attended in online mode
8	Faculty of each specialization	Dr. P.Kanta Rao	Professor, Dept. of EEE SRKR Engineering College	20/7/24
9		Prof. N.Srinivasu	Professor, Dept. of EEE SRKR Engineering College	N. Srinivasu 20-7-24
10		Dr. M.Sai Veerraju	Professor, Dept. of EEE SRKR Engineering College	20/7/24
11		Dr. M.Siva Rama Krishna	Associate Professor, Dept. of EEE SRKR Engineering College	NOT present
12				
13	Student Members	Mr. V. Kishore Kumar	20B91A02H2 B.Tech., EEE, 2020-2024 Batch	NOT present
		Ms. N. Harika	21B91A02B0 B.Tech., EEE, 2021-2025 Batch	N. Harika

Agenda:

1. To discuss and finalize the scheme and syllabi for the second year of R23 B. Tech (EEE) course curriculum.

2. Any other item with the permission of the chairman.

R23 – 2 Year – EEE Department

BOS Overall Minutes of the Meeting:

R23 – II/IV B. Tech			
Sl.no	**Suggestion**	**BOS Member**	**Resolution**
1	❖ Emf course before / parallel to DC machines and Transformers course	Dr P Shankar	In the view of other common courses, resolved to keep EMF course in 2 Year 2 Semester and include Electro mechanical energy conversion principles in DC Machines & Transformer course.
2	❖ Not overlap syllabus in PE, PED & EV courses	Dr G Kasi Viswanath Raju	Resolved to consider the suggestion while preparing 3 Year Course Syllabus.
3	❖ **Electrical circuit analysis 2:** a) Add pulse response concept after step response in unit – 1. b) Better to interchange units 4 and 5 c) Transform impedance and Transform circuits concepts are moved into unit 3 from unit 5 d) Add introduction to filters concept e) Remove reference - 1 textbook and make it as main textbook	Akula Ramesh Dr M Damodar Reddy Dr G Kasi Viswanath Raju	a) Resolved to include pulse response in Laplace Transform applications in Unit – 3. b) Suggestion accepted and units 4 & 5 are interchanged. c) Based on textbook content resolved to retain in Network function unit. d) Resolved to include introduction to filters and elementary treatment of passive filters. e) As already two text books are included, retained as reference book.
4	❖ **DC machines and transformer:** a) Make special machines as compulsory course instead of Elective. b) Include types of dc motors c) Include vector groups concept	Dr P Shankar Akula Ramesh Dr G Kasi Viswanath Raju	a) Resolved to considered while preparing 3 Year Scheme & Syllabus. b) Resolved to include types of dc motors. c) Resolved to include vector groups concept.
5	❖ **MATLAB Laboratory:** a) In Experiment 6 not use direct blocks instead of using differential equations procedure or include both methods and compare it. b) Combine experiments 1 and 2 & 7 and 8 to incorporate more other experiments. c) Use PSPICE & PSIM Softwares for transient analysis.	Akula Ramesh Dr G Kasi Viswanath Raju Dr M Damodar Reddy	a) Resolved to include suggested changes. b) In view of program complexity resolved to retain experiments as they are. c) Not considered for more focus on MATLAB programming & simulation. d) Resolved to change the title.

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	d) Change title to circuit simulation lab by modifying experiments 1, 2 and 3.		
6	❖ **Electrical measurements and instrumentation:** a) Include digital signal measurements data acquisition / logger's Concepts. b) In unit 5 better to use emerging technology, measuring devices.	Dr P Sankar Dr Bhaskar Reddy	a) Resolved to include digital storage oscilloscope. b) Resolved to include LDR, Piezo electric transducer and flow meters.
7	❖ **Electromagnetic field:** a) Review of vector calculus and coordinate system in unit – 1. b) Add applications of course.	Dr M Damodar Reddy	a) Resolved to include the suggested topics. b) Included applications on Gauss law and Ampere's law.
8	❖ **Digital electronics and logic design:** a) Add introduction to FPGA Concept. b) In units 4 and 5 covered any coding?	Akula Ramesh	a) Not considered due to syllabus burden. b) Verilog HDL coding included in units 4 & 5.
9	❖ **Digital design laboratory:** At least some experiments having both simulation and hardware.	Akula Ramesh	Resolved to include experiments on multiplexers and combinational logic circuits using both simulation and hardware.
10	❖ **Smart system laboratory:** Mention website Links for understanding various software's and measuring devices	Dr G Kasi Viswanath Raju	Resolved to include following links 1. Embedded Sensing, Actuation and Interfacing Systems, IIT Kharagpur Prof. Banibrata Mukherjee https://nptel.ac.in/courses/108105376 2. Foundations of Cyber Physical Systems, IIT Kharagpur Prof. Soumyajit Dey https://nptel.ac.in/courses/106105241 3. Introduction to internet of things, IIT Kharagpur Prof. Sudip Misra https://nptel.ac.in/courses/106105166 4. Introduction to Industry 4.0 and Industrial Internet of Things, IIT Kharagpur Prof. Sudip Misra https://nptel.ac.in/courses/106105195

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24-06-2025

CIRCULAR

This is to inform all the members that the 12th Board of Studies (BOS) meeting of the Department of EEE, SRKR Engineering College, Bhimavaram will be held on **28-06-2025 (Saturday) at 10:00 AM** in PG Simulation Laboratory of the department. All the BOS members are invited to attend the meeting. Members can attend the meeting either offline or online.

Agenda:

1. To discuss and finalize the scheme and syllabi for the Third year of R23 B. Tech (EEE) Program curriculum.
2. To discuss and finalize the courses & syllabi for the R23 B. Tech (EEE) Honors, Minors and Open Electives.
3. Any other item with the permission of the chairman.

S.R.K.R. ENGINEERING COLLEGE ★ DEPT. OF E.E.E. ★ BHIMAVARAM-534 204

Head of the Department &
BOS Chairman

Head of EEE Department
S.R.K.R. Engg.College
BHIMAVARAM –534 204

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S.R.K.R. ENGINEERING COLLEGE: CHINA AMIRAM BHIMAVARAM
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
BOS Meeting held on 28-6-2025

S. No	Category	Name of the Member	Designation & Address	Signature
1	Chairman	Dr. B.R.K. Varma	Professor & Head Dept. of EEE SRKR Engineering College China Amiram, Bhimavaram-534204	28/6/25
2	JNTUK Nominee	Dr. K. Ravindra, Ph. D	Professor Department of Electrical & Electronics Engineering UCEK, JNTUK, Kakinada.	Attended in online mode
3	Experts from other Universities	Dr. Ravikumar Bhimasingu, Ph. D (IISc Bangalore)	Professor Department of Electrical Engineering Indian Institute of Technology Hyderabad – 502284.	Attended in online mode
		Dr. Vuddanti Sandeep, Ph. D (IIT Delhi)	Associate Professor Department of Electrical Engineering National Institute of Technology Andhra Pradesh – 534101.	Attended in online mode
4	Industry Experts	Dr. Adapa Anil, Ph. D (IISc Bangalore)	Director of Electrical Engineering Ohmium Bangalore. Karnataka – 560078.	Attended in online mode
5	Alumni	Dr. P.V.N Mohan Krishna, Ph. D (IIT Bhubaneswar)	Market Research and Analyst Hitachi Energy, Bhorukha Tech Park, Mahadevapura, Banglore. Karnataka – 560078.	Attended in online mode
6	Faculty of Department	Prof. N. Srinivasu	Professor	
		Dr. M. Saiveerraju	Professor	
		Dr. P. Kantharao	Professor	
		Dr. G. Veeranna	Associate Professor	
		Dr. G. Harish Kumar Varma	Associate Professor	
		Dr. G Pavan Kumar	Associate Professor	
		Dr. Md. Azahar Ahmed	Associate Professor	
		Dr. G. Kusuma	Associate Professor	
		Dr. G. Syamnaresh	Associate Professor	
		Dr. Ch. Murali	Assistant Professor	
		Sri K. Kiran Kumar	Assistant Professor	
		Dr. K. Swetha	Assistant Professor	

Dr. K.S.S.Prasadraju	Assistant Professor	
Dr. I. Swetha Monica	Assistant Professor	
Dr. A. Kumarraja	Assistant Professor	
Sri V. Srinivas	Assistant Professor	OD
Sri Y. Bhanuchandar	Assistant Professor	
Sri Sk. Nazeer	Assistant Professor	
Dr. N.V.A. Bhavani	Assistant Professor	OD
Sri B. Venkata Krishna	Assistant Professor	
Dr. D.A.K. Rao	Assistant Professor	
Sri S. Rajasekhar Reddy	Assistant Professor	OD
Dr B S S Santosh	Assistant Professor	24/6/15
Dr. P.Jagadeesh	Assistant Professor	
Dr. Ch.Durga Prasad	Assistant Professor	
Sri M.E.C.Vidyasagar	Assistant Professor	OD
Sri P.Sailesh Babu	Assistant Professor	OD
Sri E.Suresh	Assistant Professor	
Smt Bh.Sudharani	Assistant Professor	
Sri A.Hemanth Kumar Raju	Assistant Professor	
Sri K.Pavan Kumar	Assistant Professor	
Sri P.S.P.R.Swamy	Assistant Professor	
Sri D.Rajesh	Assistant Professor	
Ms. A M S V Sushma	Assistant Professor	OD
Sri. Y V V S Narayana	Assistant Professor	
Smt. P. Udaya Bhanu	Assistant Professor	
Sri. I Kranthi Kumar	Assistant Professor	OD
Sri. B. Eswar Rao	Assistant Professor	

BOS: R23 - (2025-2026) Overall Minutes of the Meeting:

Sl. No	Suggestion	BOS Member	Resolution
1	Any IOT Laboratory in Third year?	Dr. K. Ravindra	No Specific IoT course in 3 year, but Smart System Laboratory & Tinkering Laboratory courses are included in 2 & 3 Year.
2	Any course related to "Thermal Dynamics"?	Dr. Adapa Anil	NO course included in the curriculum related to Thermal Dynamics.
3	Courses named as "PSOC & PSP" in different groups.	Dr. P.V.N Mohan Krishna	Resolved to consider the suggestion while preparing 4th Year Course Syllabus.
4	**Transmission & distribution Systems**		
	Add 3rd edition in Text book - 2	Dr. V. Sandeep	Resolved to include suggested change.
5	**Power Electronics**		
	a) Instead of "choppers" change the title to "DC – DC Converters". b) Change "step – up" to "boost" and "Step – down" to "buck" in Unit – 2. c) Introduce 1-phase diode bridge rectifier with filter in Unit - 2.	Dr. K. Ravindra	a, b) Resolved to change the titles. c) Resolved to include 1-phase diode bridge rectifier with filter in Unit – 2.
6	**Solar and Wind Energy Systems**		
	Add "Renewable and Efficient Electric Power Systems" by Gil Masters.	Dr. K. Ravindra	Resolved to include suggested book.
7	**Energy Storage Systems**		
	Include "Sodium ion Batteries" in Unit – 2.	Dr. Anil Adapa	Resolved to include topic "Sodium ion Batteries" in Unit – 2.
8	**Power Electronics Laboratory**		
	a) Include R-L load in Experiment – 7 b) Include generation of PWM using FPGA c) Add variable R – load for CCM & DCM for Buck converter and Boost converter.	Dr. K. Ravindra	a) Resolved to include suggested change. b) Resolved to include suggested change in the next curriculum revision. c) Resolved to include suggested change in experiments 4 & 5.
	a) Instead of Mid – point cyclo converter use H – bridge multilevel inverter. b) 1-phase half bridge inverter using unipolar & bipolar as Experiment – 9.	Dr. Anil Adapa	a) Resolved to include suggested change. b) Resolved to include suggested change in the next curriculum revision.
9	**Solar Energy Systems Laboratory**		
	Add simulation for Solar MPPT Experiment	Dr. K. Ravindra	Resolved to include suggested change.
10	**Tinkering Laboratory**		
	a) Make Tinkering Lab advanced to smart system lab. b) Make experiments as applications to electrical engineering.	Dr. V. Sandeep	a) Developed Experiments advanced to smart system lab. b) Experiments 1-6, 8 were added as applications to electrical engineering.

	a) Make Tinkering Lab with Three Themes: computer vision, IoT and PCB Prototype. b) Add experiment related to PCB prototype.	Dr. P.V.N Mohan Krishna	a) Resolved to include suggested change. b) Experiments 1,2 & 4 were added as a part of PCB Prototype.
11	**Microprocessors and Micro Controllers**		
	Remove 8051 Architecture and go for any micro controller only with C - programming.	Dr. Anil Adapa	ARM Controller is Included instead of 8051 Microcontroller.
	Replace Unit – 4 and Unit - 5 with ARM processor instead of 8051 Microcontroller.	Dr. K. Ravindra	Unit – 4 and Unit - 5 are replaced with ARM processor & removed 8051 Microcontroller.
12	**Control Systems**		
	No need to include feedback characteristics in Unit – I.	Dr. K. Ravindra	Resolved to include suggested change.
	If possible, remove observability and controllability in Unit – V.	Dr. Anil Adapa	Observability & controllability concepts are retained in view of GATE examination and advanced courses.
	Add Katsuhiko Ogata, "Modern Control Engineering" as main textbook.	Dr. K. Ravindra	Retained in the reference book.
13	**Power Electronics for Renewable Energy**		
	Half bridge and Full bridge with Transformer isolation should be added instead of Flyback converter.	Dr. Anil Adapa	Full bridge with Transformer isolation added in Unit – 4 instead of Flyback converter.
14	**Electrical Distribution Systems**		
	One or two components' data sheet required for Protective device characteristics.	Dr. Anil Adapa	Type K & Type T fuse links Datasheets added.
16	**Unmanned Aerial Vehicles** Check syllabus from IIT Kanpur	Dr. P.V.N Mohan Krishna	Resolved to include suggested change in the next curriculum revision
17	**Microprocessors & Microcontrollers Laboratory**		
	Add "ARM Cortex" experiments instead of 8051 micro controllers.	Dr. V. Sandeep	ARM Cortex experiments are included.
18	**Control Systems Laboratory**		
	Add "Transfer function of DC Servo Motor" experiment.	Dr. K. Ravindra	Included Torque Speed characteristics of DC Motor
20	**Electrical System Simulation Laboratory**		
	Add "Electrical Transients in Power Systems" by Allan Greenwood as Reference text book.	Dr. P.V.N Mohan Krishna	Resolved to include suggested book.
21	**Semiconductor Devices**		
	Add fundamentals of "wideband gap devices" in Unit – 5.	Dr. Anil Adapa	Resolved to include suggestion in Unit – 5.
22	**Power Conversion Systems & Battery Storage**		
	Change to CC & CV modes rather than CCCV	Dr. Anil Adapa	Resolved to include suggestion in Unit – 5.

23	**Advanced Power Electronics**		
	SEPIC and ZETA converters should be removed in Unit – 1.	Dr. Anil Adapa	Resolved to remove.
	Remove canonical circuit model & Transfer function model and add averaged switch model technique in Unit – 2.	Dr. K. Ravindra	Average switch model and Averaged state space model are included in Unit – 2.
	Remove Half bridge & Full Bridge converters in Unit -3.	Dr. Anil Adapa	Resolved to include.
	a. 6 & 12 pulse are sufficient and remove 18& 24 pulse in Unit – 4. b. Remove applications of modulation techniques to inverters and add THI – SPWM technique for 3-phase 2 level inverters & carrier - based schemes for 3-phase multilevel inverters.	Dr. K. Ravindra	a) Resolved to include. b) Applications of modulation techniques to inverters are removed and THI – SPWM technique & carrier - based schemes for 3-phase multilevel inverters are included.
24	**MATLAB programming for engineering applications**		
	Add any application related concepts in Unit – 2.	Dr. Anil Adapa	Resolved to include.

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